



Development of an Integrated Forecasting System for Inventory Management in Oil Depots

GENERAL SCOPE OF THE PROJECT

I- Problematic

II- Objectives

III. Architecture





I- Problematic



- Confronted with an immense data stream, the complexity arises from the absence of real-time visibility and the lack of a forecasting system, hindering swift decision-making and proactive inventory management. This challenge is evident in the struggle to predict demand, optimize stock levels, and proficiently handle intricate errors that may emerge throughout the process.
- The absence of an effective logistics strategy for transporting petroleum products, from the production site to the depot and distribution.



II- Objectives

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- Implement a forecasting system for oil depots to avoid stock-outs, optimize ordering processes, analyze tank behavior across various parameters, in order to guarantee customer satisfaction.
 - Study the essential operational and logistical criteria.
 - Generate a reporting dashboard to clarify the results.



Forecasting aims

The development of a forecasting system for stock management, serving to:

- Prevent stockouts to maintain customer satisfaction
- Ensure product availability in stocks to meet customer demand
- Determine the behavior of tanks using various parameters (volume, meter, injector)
- Improve order processes to determine optimal quantities for replenishment, considering various criteria (Density, pressure, temperature, etc)

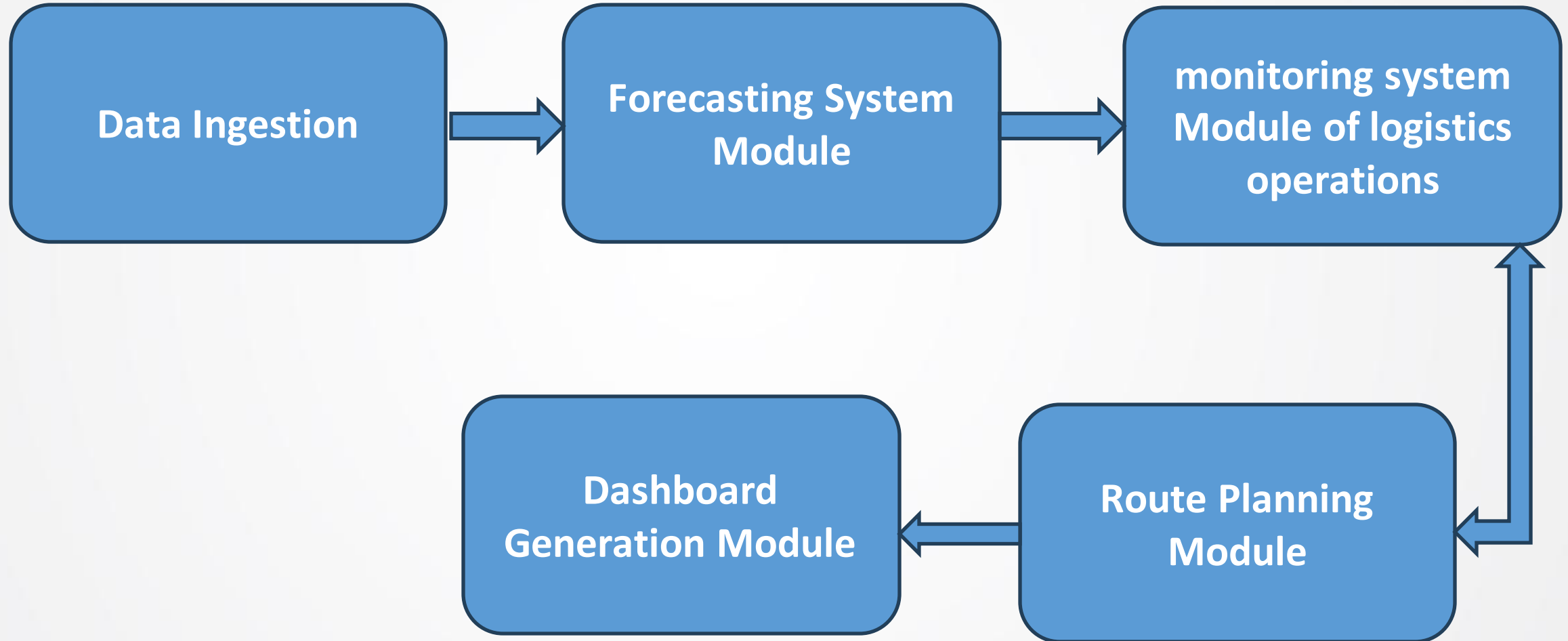
Logistics aims

- Optimization of logistics chains for efficient transport of petroleum products from production site to depot and distribution.



III- Solution Architecture

► Conceptuel Diagram





KeyWords



- Terminal Automation System (TAS): <https://pgsintl.com/brochure>
- Bulk movement
- Bay querying
- Tang inventory



► ***GOOD LUCK ☺***